

Graphical relationships and Inverse Functions — Problem Set

Warm-up

1. Find the equation of the inverse function $f^{-1}(x)$ for the linear function $f(x) = 3x - 5$.
2. If $f(x) = \frac{1}{x+2}$, write down the equation of the reciprocal function $y = \frac{1}{f(x)}$.
3. Given the point $(3, -1)$ lies on the graph of a one-to-one function $y = f(x)$, state the coordinates of the corresponding point on the graph of $y = f^{-1}(x)$.
4. Sketch the graph of $y = |x^2 - 4|$ by transforming the graph of $y = x^2 - 4$.
5. Sketch the graph of $y = (|x-2|)(|x+1|)(|x|)$ by transforming the graph of $y = (x-2)(x+1)(x)$.
6. Find the domain of the function $f(x) = \sqrt{x-3}$.
7. Find the equation of the inverse function $f^{-1}(x)$ for the linear function $f(x) = 4x + 7$.
8. If $f(x) = x - 3$, write down the equation of the reciprocal function $y = \frac{1}{f(x)}$ and state its vertical asymptote.
9. Given that the point $(-2, 5)$ lies on the graph of a one-to-one function $y = f(x)$, state the coordinates of the corresponding point on the graph of $y = f^{-1}(x)$.
10. Find the domain of the function $f(x) = \sqrt{6-2x}$.
11. For the function $f(x) = 2x - 3$, find the explicit expression for the composite function $f(f(x))$.
12. A function $f(x)$ has a domain of $[-1, 4)$ and a range of $(2, 7]$. State the domain and range of its inverse function $f^{-1}(x)$.
13. Sketch the graph of $y = |x^2 - 9|$ by transforming the graph of $y = x^2 - 9$, clearly showing all intercepts.
14. State the equations of all vertical and horizontal asymptotes for the rational function $y = \frac{1}{x^2-4}$.
15. Sketch the graph of $y = |x - 3| - 1$, indicating the coordinates of the vertex and any axis intercepts.
16. Determine whether the function $f(x) = x^2 + 4$ defined for the restricted domain $x \geq 0$ possesses a unique inverse function.

Skill-building

1. Consider the function $f(x) = \frac{1-x}{1+x}$ for $x \neq -a$. Show that $f(f(x)) = x$ for all x in the domain, and explicitly state what this implies about the relationship between $f(x)$ and $f^{-1}(x)$.
2. Sketch the graph of $y = \frac{1}{f(|x|)}$ where $f(x) = (x-2)(x-4)$, clearly indicating all vertical asymptotes and horizontal asymptotes

3. Let $f(x) = x + \sqrt{x^2 + 1}$ for all $x \in \mathbb{R}$.

1. Find an explicit expression for the inverse function $y = f^{-1}(x)$.

4. The function $f(x) = x^4 - 4x^2 + 3$ is restricted to a domain $x \leq k$ such that it possesses a unique inverse function $f^{-1}(x)$. Find the maximum possible value of k .

5. For the function $f(x) = |x - 1| - |x + 2|$:

1. Sketch the graph of $y = f(x)$.

2. Sketch the graph of $y = \frac{1}{f(x)}$, showing all features.

6. Let $f(x) = \frac{x}{\sqrt{1-x^2}}$ for $-1 < x < 1$. Find the algebraic equation for $f^{-1}(x)$ and state its domain and range.

7. A function is defined piecewise by:

$$f(x) = \begin{cases} -x^2 - 2x & \text{for } x < -1 \\ |x| & \text{for } x \geq -1 \end{cases}$$

1. Sketch $y = f(x)$.

2. Sketch $y = f(x) + f(-x)$, showing all intercepts and sharp edges.

8. The function $f(x) = x^2 - 6x + 8$ is restricted to a domain $x \leq k$ such that it possesses a unique inverse function $f^{-1}(x)$. Find the maximum possible value of k .

9. Let $f(x) = 2x + \sqrt{4x^2 + 1}$ for all $x \in \mathbb{R}$. Find an explicit algebraic expression for the inverse function $y = f^{-1}(x)$.

10. Sketch the graph of $y = \frac{1}{f(x)}$ where $f(x) = x^2 - 4x + 3$, clearly indicating all vertical asymptotes, axial intercepts, and the coordinates of the local maximum turning point.

11. Let $f(x) = \frac{x}{\sqrt{1+x^2}}$ for all $x \in \mathbb{R}$. Find the algebraic equation for $f^{-1}(x)$ and state its domain and range.

12. For the function $f(x) = |x - 2| - |x + 3|$:

1. Sketch the graph of $y = f(x)$.

2. Sketch the graph of $y = \frac{1}{f(x)}$, showing all asymptotes and key features.

13. A function is defined piecewise by:

$$f(x) = \begin{cases} -x^2 - 4x & \text{for } x < -2 \\ |x| & \text{for } x \geq -2 \end{cases}$$

1. Sketch $y = f(x)$.

2. Sketch $y = f(x) + f(-x)$, clearly showing all key features and sharp edges.

14. Consider the function $f(x) = \frac{1}{3}(x - 4)^2 + 2$.

1. Sketch the parabola $y = f(x)$, showing clearly the coordinates of the vertex.

2. Write down the largest domain containing the value $x = 2026$ for which the function has a well-defined inverse function $f^{-1}(x)$.

3. For the domain found in part (b), find the explicit algebraic expression for $f^{-1}(x)$.

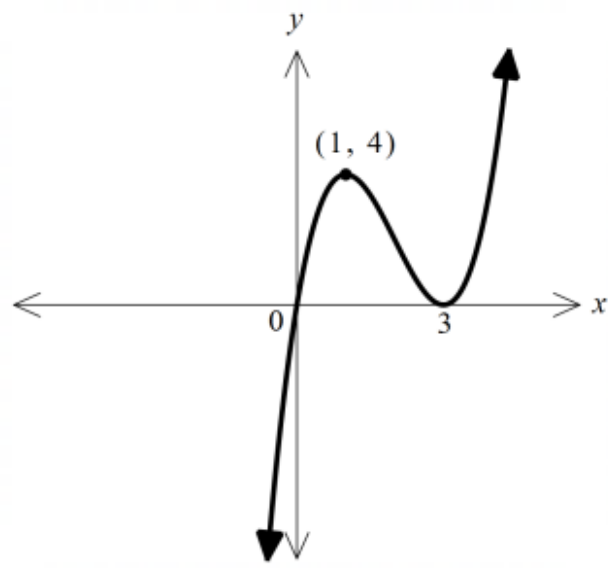
Easier Exam Questions

1. (Baulkham Hills 2020 Q12b)

The diagram shows the graph of a function $y = f(x)$.

(i) Draw the graph of $y = \frac{1}{f(x)}$ showing all important features.

3

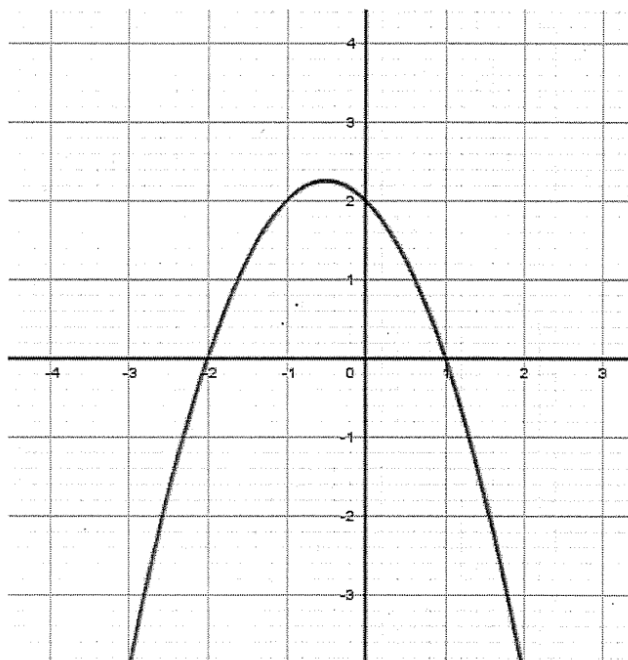


(ii) Draw the graph of $y = f(|x|)$ showing all important features.

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2. (Girraween (2022) Q6c)

The diagram shows the graph of the function $f(x) = (x + 2)(1 - x)$.



Sketch the following:

(i) $y = \frac{1}{f(x)}$ **3**

(ii) $y = |f(x)|$ **2**

(iii) $y = f(|x|)$ **2**

3. (James Ruse (2022) Q9)

For which of the following conditions does the graph of

$$y = \frac{1}{ax^2 + bx + a} \text{ for } a \in \mathbb{R} \setminus \{0\}$$

have exactly two asymptotes?

(A) $-2 < \frac{a}{b} < 2$

(B) $-2 \leq \frac{a}{b} \leq 2$

(C) $-1 < \frac{b}{a} < 1$

(D) $\frac{a}{b} = \frac{1}{2}$

4. (Caringbah (2019) Q13b)

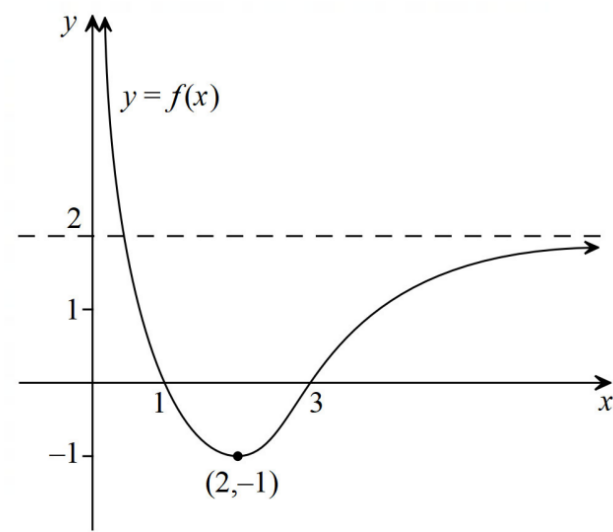
Find the equation of the inverse $f^{-1}(x)$ for $f(x) = \frac{1-x}{3+x}$.

2

Harder Exam Questions

1. (Baulkham Hills (2022) Q12a)

Consider the graph of $y = f(x)$ below.



On separate axes, sketch each of the following, showing all important features.

3

$$y = \frac{1}{f(|x|)}$$

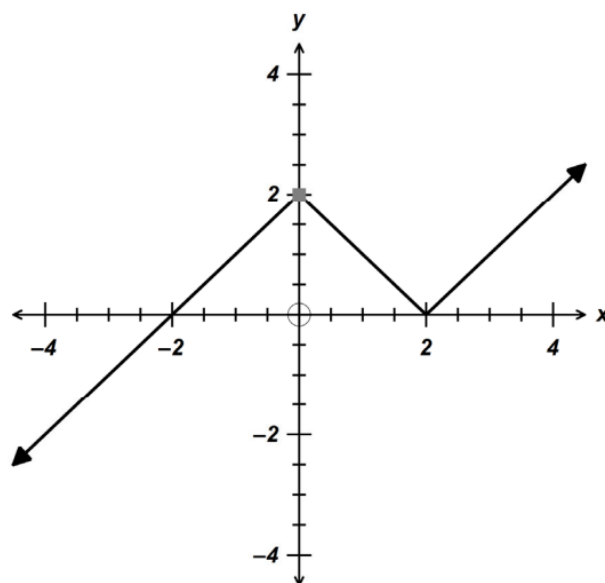
2. (Blacktown Boys (2023) Q12(e))

Sketch the graph of $f(x) = -|x|^2 + 4|x| - 3$, showing all key features.

2

3. (Presbyterian (2022) Q12d)

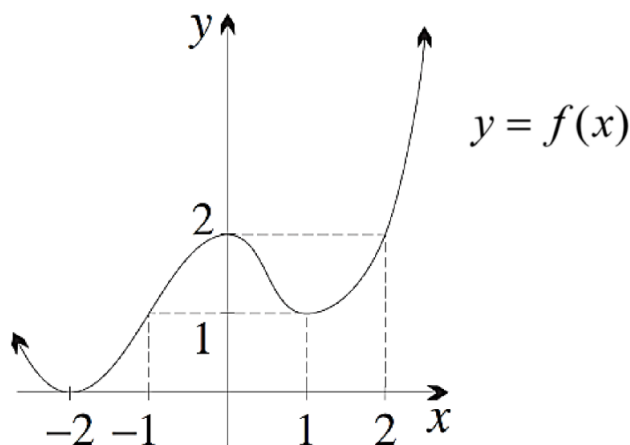
The graph below shows $y = f(x)$ where $f(x) = x + 2$ for $x < 0$ and $f(x) = |x - 2|$ for $x \geq 0$



Sketch the graph of $y = f(x) + f(-x) + 1$, carefully showing any intercepts on the co-ordinate axes. **3**

4. (Fort St (2022) Q4)

Given the graph of the quartic $y = f(x)$ below, which of the following restrictions on the domain would allow the function $f(x)$ to have an inverse function $y = f^{-1}(x)$?



- (A) $[-2, 1]$
- (B) $[-2, 2]$
- (C) $[0, \infty)$
- (D) $[-2, 0]$

5. (Presbyterian (2023) Q1)

The function, $f(x)$, is one-to-one and has domain $[0, 5)$ and range $(-2, 5]$. Which of the following having correctly describes the domain and range of its inverse function, $f^{-1}(x)$?

	Domain	Range
(A)	$(-2, 5]$	$[0, 5)$
(B)	$[-2, 5)$	$(0, 5]$
(C)	$[0, 5)$	$(-2, 5]$
(D)	$(0, 5]$	$[-2, 5)$

6. (Sydney Boys (2019) Q15d)

The function $f(x)$ is defined by $f(x) = \frac{1-x}{1+x}, x \neq -1$ Given that $f^2(x) = f(f(x))$, by first finding the simplest expression for $f^2(x)$, show that $f^{2020}(x) = x, x \neq -1$. **2**

7. (Sydney Girls (2022) Q4a(i))

(a) Consider the function $f(x) = \frac{1}{2}(x-3)^2 + 4$.

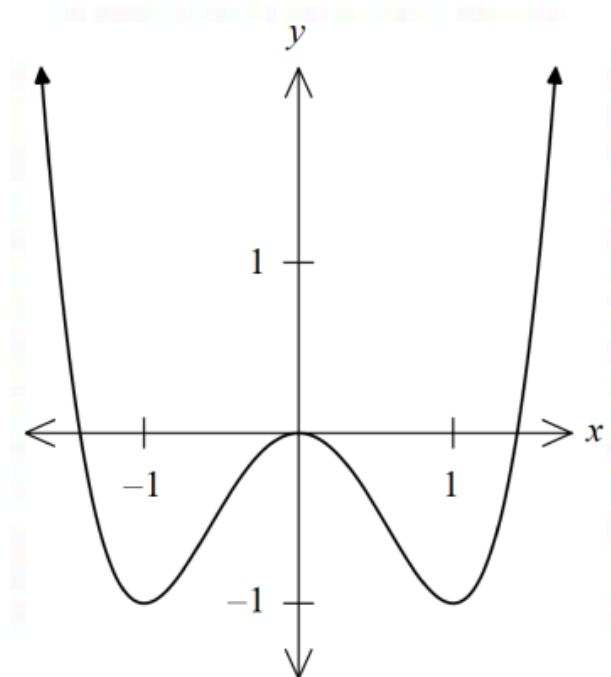
(i) Sketch the parabola $y = f(x)$, showing clearly the coordinates of the vertex. Hence write down the largest domain containing the value $x = 2022$, for which the function has an inverse function $f^{-1}(x)$. **2**

(ii) For the domain found in part (i), find $f^{-1}(x)$. **2**

(iii) Let a be a real number NOT IN the domain found in part (i). By considering the symmetry of the parabola, or otherwise, find $f^{-1}(f(a))$ **2**

8. (James Ruse (2024) Q5)

The function $f(x) = -\sqrt{1 - \sqrt{1+x}}$ has an inverse $f^{-1}(x)$. The graph of $y = f^{-1}(x)$ forms part of the curve $y = x^4 - 2x^2$.



How many points do the graphs of $y = f(x)$ and $y = f^{-1}(x)$ have in common?

(A) 1

(B) 2

(C) 3

(D) 4